Infrared Line Tracking Data Collection

1. Experiment Purpose

The Lastbit car is equipped with a four-channel infrared line tracking module from the factory. This lesson will guide you through learning how to read the data returned by the four-channel line tracking module and display the data on the Micro:bit LED matrix.

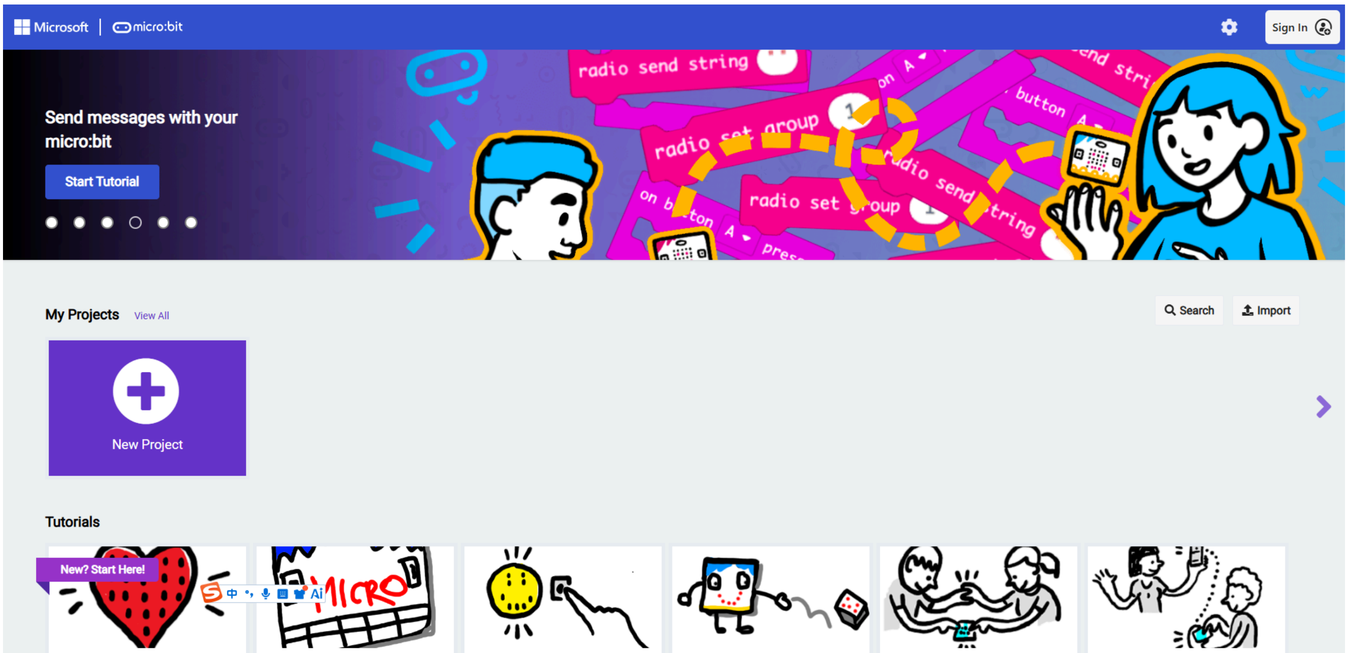
The infrared line tracking module consists of four pairs of infrared transmitting tubes and receiving tubes. When operating, the infrared transmitting tubes emit infrared light. When the infrared light is reflected by the surface of an object, the receiving tube outputs high or low level signals based on the intensity of the reflected light. Black objects absorb infrared light and reflect weakly, so the module outputs a low level. White objects reflect strongly, so the module outputs a high level. By detecting the level status of the four channels, the position of the car on the black line track can be determined.

The four-channel line tracking module interface has been connected from the factory, and the interface is designed with anti-misinsertion features. You don't need to worry about damaging components when using the factory-provided cables.

1. Before starting the experiment, connect the Micro:bit mainboard to your computer using a Micro USB data cable. At this time, the computer will display the MICROBIT drive letter.

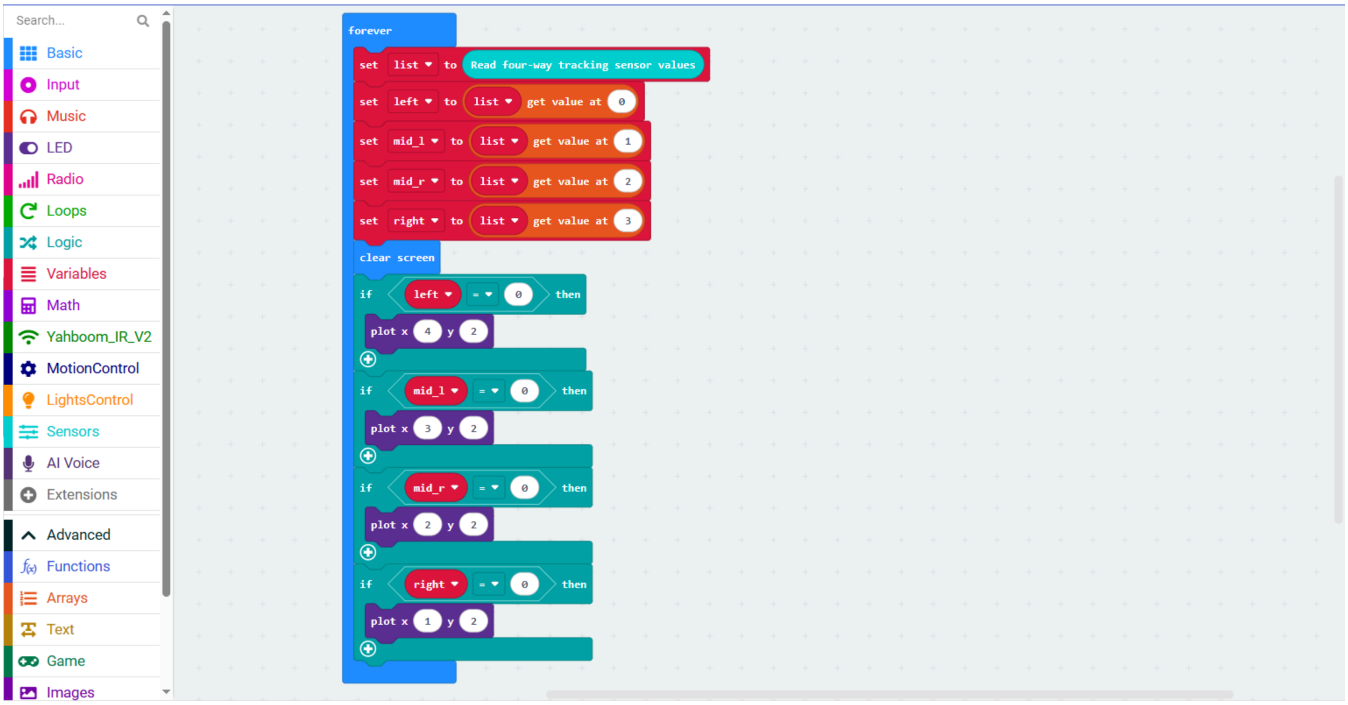
2. Open MakeCode using the link below.

   [Microsoft MakeCode for micro:bit]



3. Drag `3.Infrared_Line_Tracking_Data_Collection.hex` to the new project page. MakeCode will automatically import and open the project.

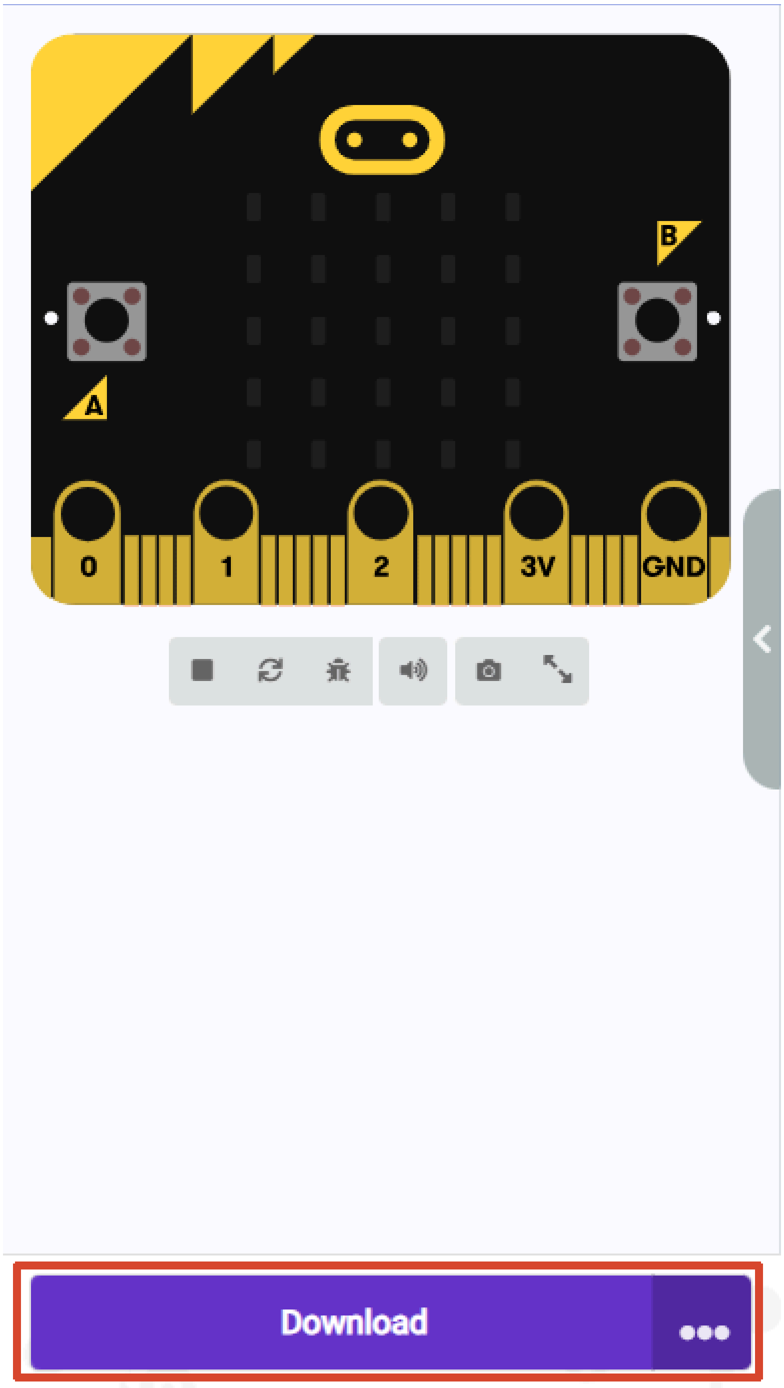
3. Experiment Source Code



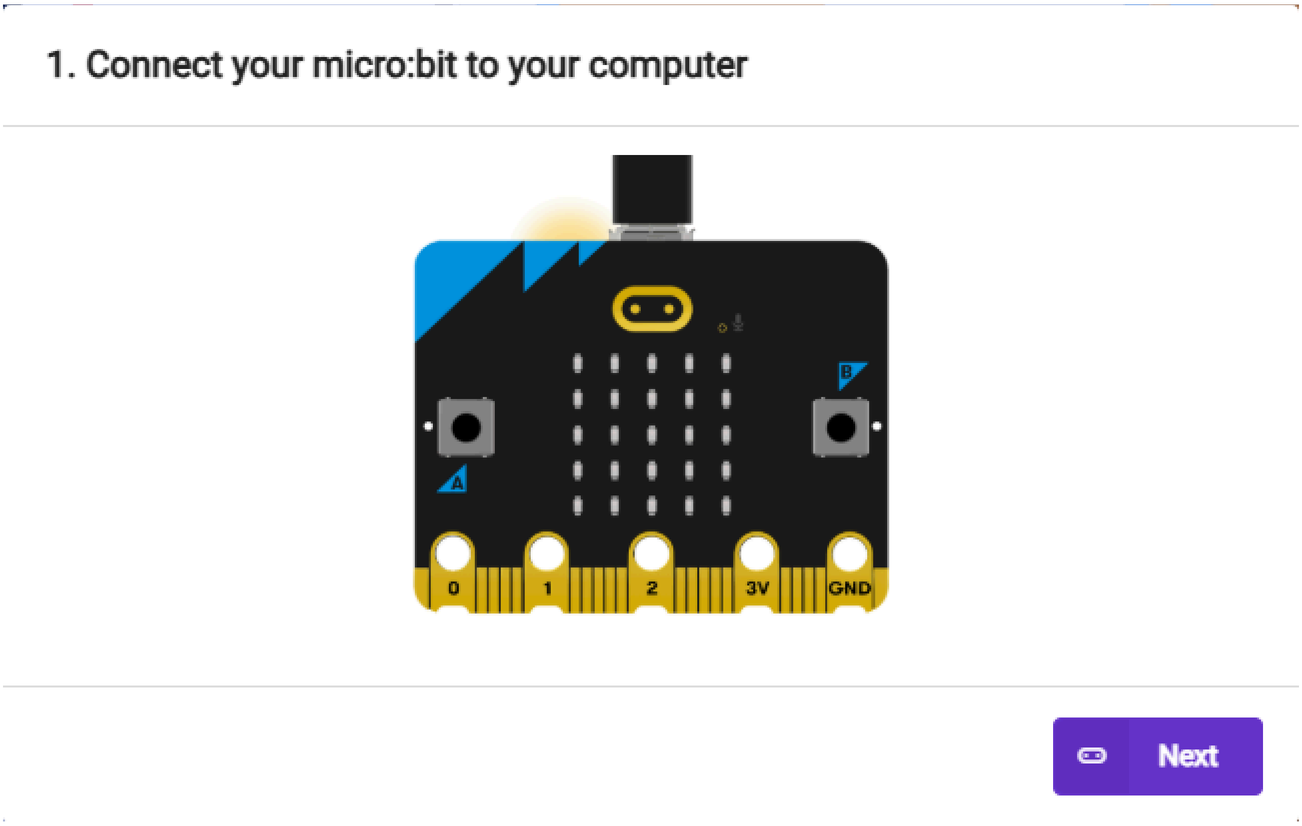
The building blocks used in this lesson are located under the `Sensors` category.

4. Program Download

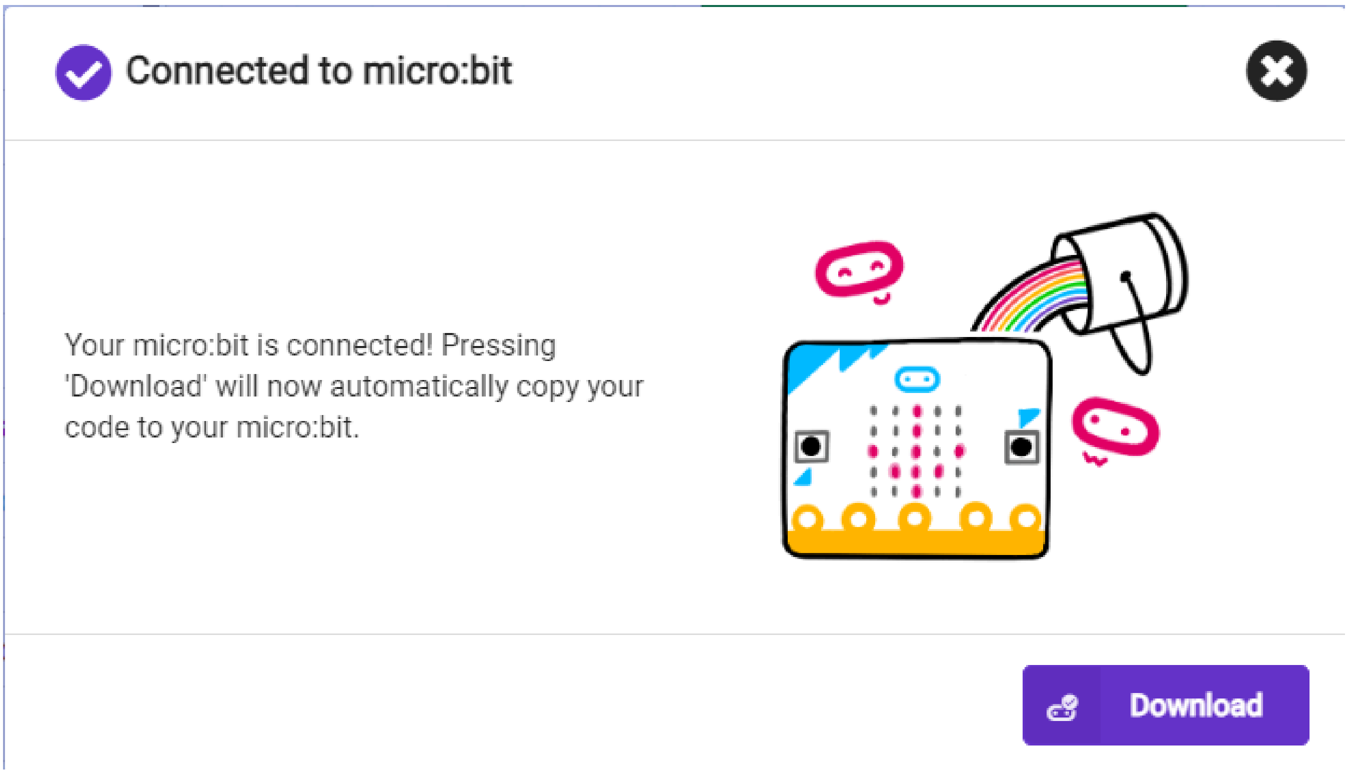
1. Click the download button at the bottom left to download the program to the Micro:bit.



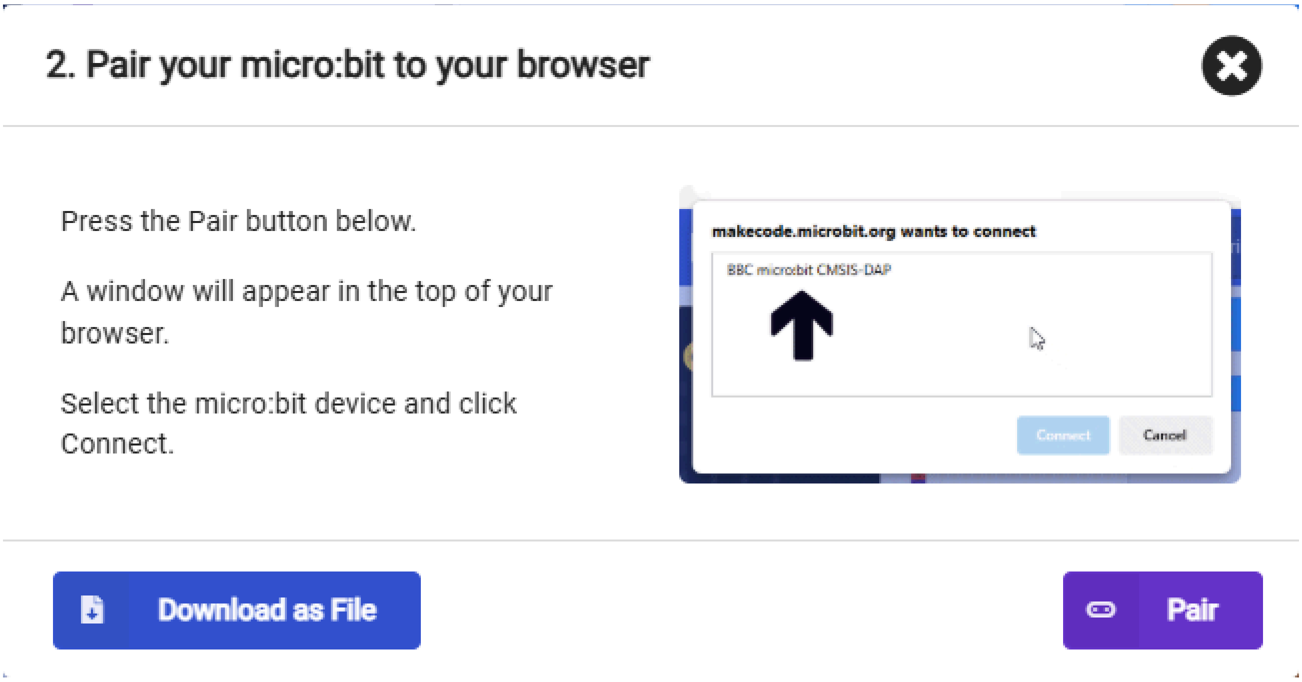
2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please connect the data cable to the Microbit first. If already connected, click [Next] here.



3. If the Microbit is detected as properly connected, the following interface will be displayed. At this point, you can click [Download].

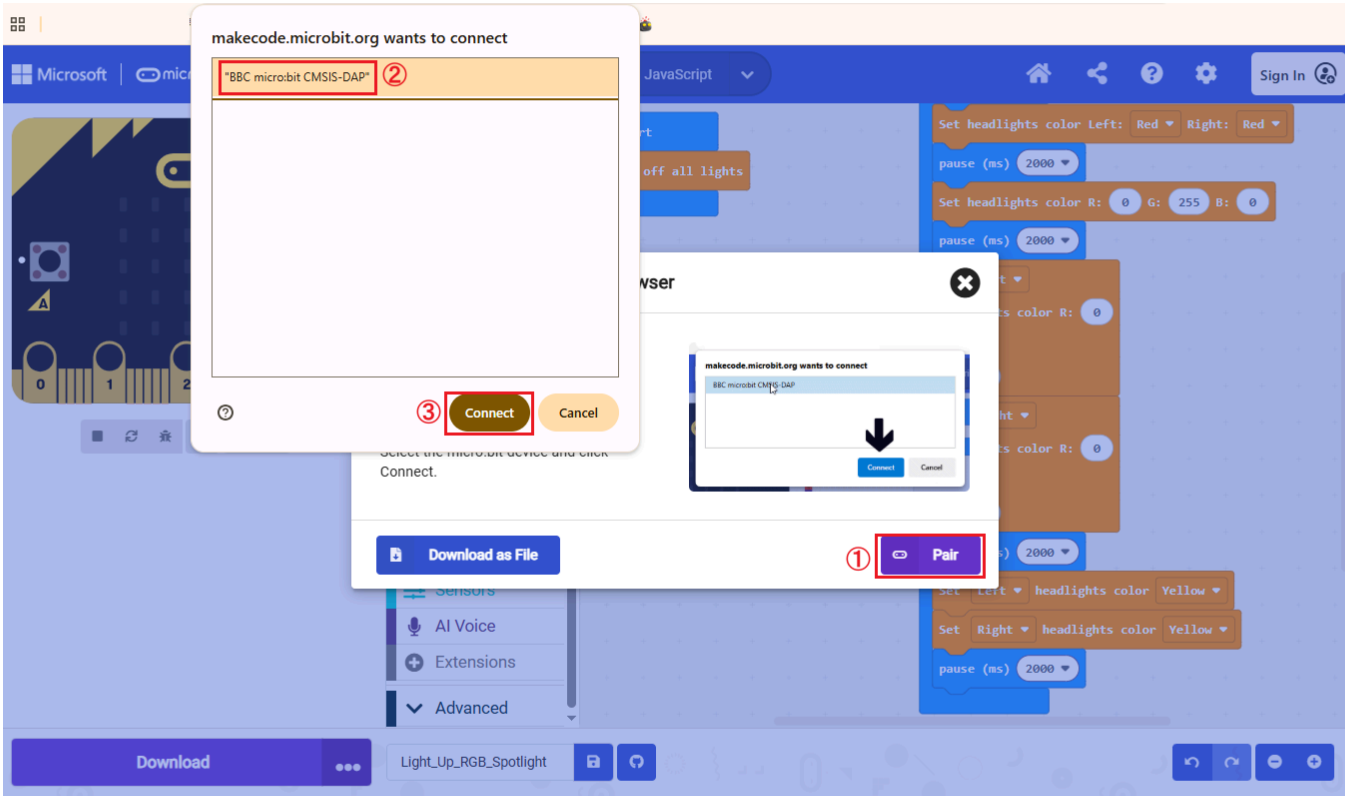


If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. You can choose between two methods: [Download as File] and [Pair].

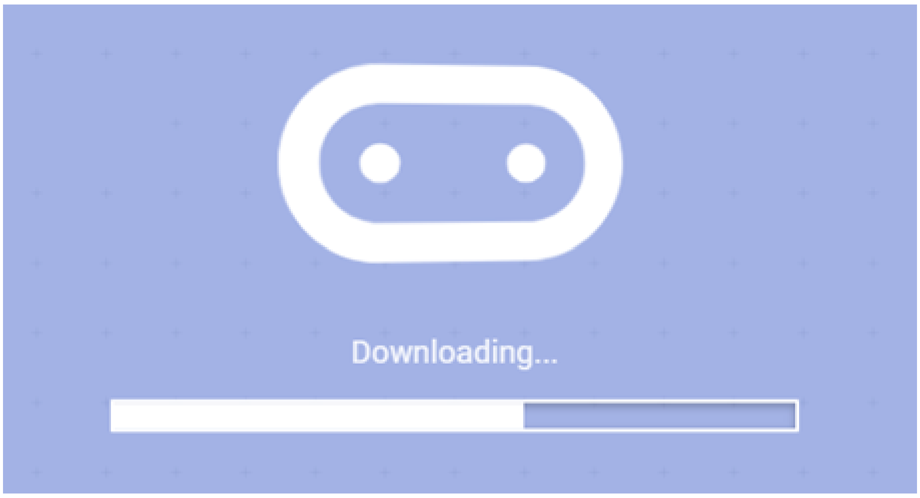


When selecting [Download as File], the system will download the currently opened program as `microbit-project-name.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive letter to complete the download.

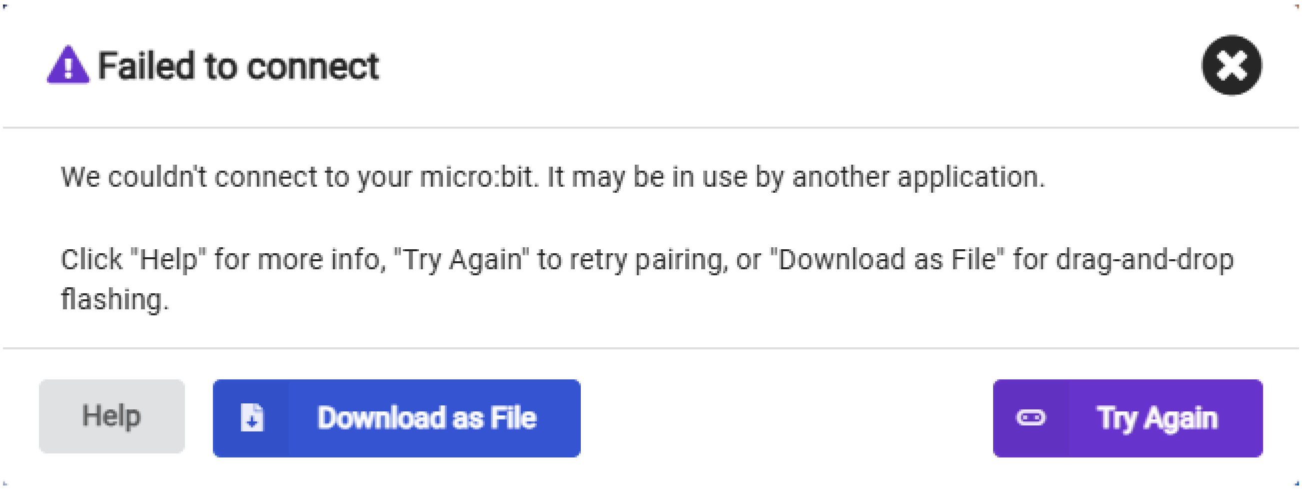
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC mico:bit CMSIS:DAP`.



4. When the following interface appears, it means the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



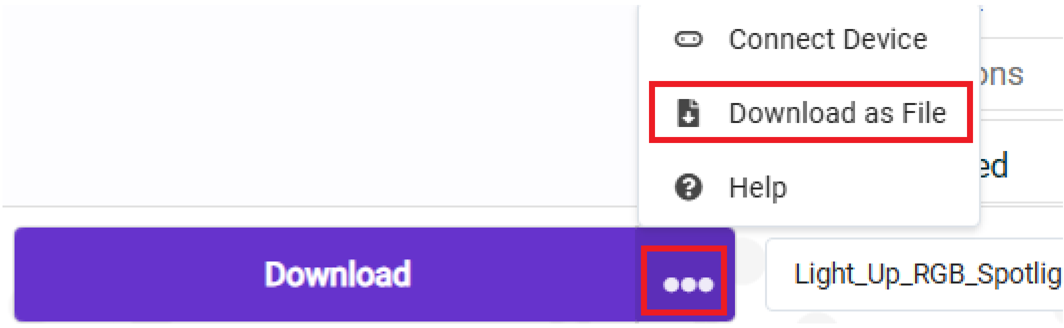
If multiple attempts continue to fail, please directly select [Download as File], save the current program as a file, and directly copy or drag the program to the corresponding MICROBIT drive letter to complete the download.



If you want to save the file directly, we provide two methods:

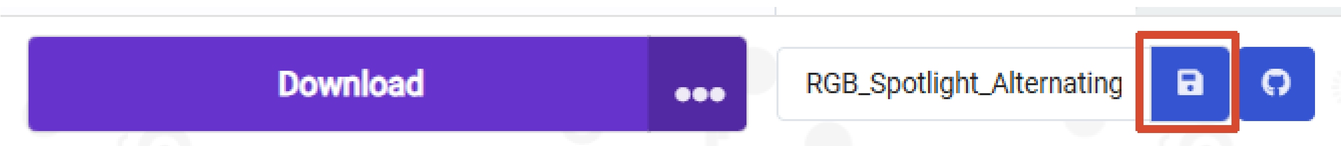
Method 1:

Click the [...] next to download, then select [Download as File]



Method 2:

Click [Save] after the [Project Name Edit Field].



5. Experiment Results

After flashing, turn on the switch (flip the switch to the ON position). Lift the car up and away from the table. You can see all four LEDs light up.

At this time, the module returns data 0. Place your finger under one of the sensors, and the corresponding LED will turn off. At this point, the returned data is 1. You can test the four channels one by one. The middle row of LEDs on the Micro:bit will light up or turn off following the indicator lights of the four line tracking modules. If one channel's sensor is always in the LED-on state, you can try rotating the potentiometer knob on top of the module to adjust the sensitivity.